

Notes: Caoui (RES, 2023)
Estimating the Costs of Standardization: Evidence from the Movie Industry

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1 Big picture

- **Question.** Can an industry that is already coordinated on an old technology standard inefficiently delay switching to a superior, incompatible new standard? If so, how large are the welfare costs of that delay?
- **Setting.** The paper studies the switch of the French movie exhibition and distribution industries from 35mm film to digital cinema between 2005 and 2013.
- **Why the setting is ideal.** The industry is a hardware-software system. Theatres need digital projectors to screen digital movies; distributors need a sufficiently large digital screen base to justify releasing movies in digital. This creates indirect network effects.
- **Main friction.** The new standard is incompatible with the old one. A theatre adopting digital projectors benefits distributors and other theatres by increasing the value of digital distribution, but the theatre does not internalize all of those benefits. Upstream distributors can also delay because each distributor wants others to move first.
- **Empirical strategy.** The paper builds and estimates a dynamic game of screen-level digital projector adoption by theatres and reduced-form digital movie supply by distributors.
- **Computational innovation.** Instead of solving a full Markov perfect equilibrium with hundreds of theatres, the paper uses a non-stationary oblivious equilibrium approximation: theatres condition on their own type and state plus time, while taking the aggregate diffusion path as deterministic.
- **Main quantitative finding.** Market forces delayed the transition relative to efficient benchmarks. Theatre surplus under the market outcome is 76.18 million 2010 euros; it is 82.25 million euros under the planner benchmark and 90.65 million euros under full coordination.
- **Decomposition.** Excess inertia reduces surplus by 16% relative to full coordination. Downstream adoption externalities across theatres explain 41% of this surplus loss, and the remaining 59% is attributed to upstream distributor coordination frictions.
- **Policy results.** A modest blanket subsidy of 25% raises net industry profits by 7.93%; a 25% subsidy targeted to miniplexes raises profits by 7.65% at a much lower fiscal cost. Larger subsidies accelerate adoption too much and lower net profits.
- **Economic takeaway.** Standardization creates coordination benefits but can also create lock-in. The welfare cost is not merely that adoption is slow; it is that private adopters do not internalize the value they create for complementary goods producers and other users.

2 Incremental contribution of the paper

2.1 Contribution relative to the theory of standards and network effects

- The classic theory of technology standards emphasizes that network effects can create *excess inertia*: firms may remain too long with an old standard because unilateral switching is unattractive.
- The paper converts that theoretical idea into an estimable quantitative object. It does not only say that inertia exists; it measures the magnitude of the surplus loss and decomposes the sources of delay.
- The paper distinguishes two kinds of inertia: downstream adoption externalities among theatres and upstream coordination failure among distributors.

Interpretation: The key incremental move is from qualitative theory to a structural welfare decomposition. The paper asks not merely whether diffusion is slow, but whether it is inefficiently slow and which market side is responsible.

2.2 Contribution relative to empirical technology-adoption papers

- Many empirical adoption papers study diffusion curves or estimate reduced-form effects of costs, demand, or peers.
- This paper combines detailed adoption data with a dynamic industry model, so it can simulate counterfactual adoption paths, efficient benchmarks, and policy remedies.
- The screen-level adoption decision is important because theatres often convert only part of their stock of screens, so adoption is not a binary firm-level event.

Interpretation: The empirical contribution is to measure adoption at the level where indivisibility and partial conversion matter. The distinction between inter-firm and intra-firm adoption is central because it reveals gradual adoption within theatres rather than a simple one-shot decision.

2.3 Contribution relative to dynamic games

- Full Markov perfect equilibrium estimation is infeasible because the French theatre industry contains hundreds of theatres and industry-level network effects.
- The paper uses a non-stationary oblivious equilibrium concept to make estimation and counterfactuals feasible.
- This approach retains the strategic economic channel that matters most - the aggregate digital installed base - while avoiding the complete industry state vector.

Interpretation: The methodological contribution is to show how a large dynamic adoption game with network effects can be estimated using an equilibrium approximation disciplined by observed diffusion paths.

2.4 Contribution relative to policy analysis

- The paper evaluates blanket subsidies, targeted subsidies, and coordination/mandate-style policies.
- It shows that more subsidy is not always better: overly aggressive subsidies speed up adoption before benefits justify costs.
- Targeting small theatres can achieve similar surplus improvements at lower fiscal cost than subsidizing all adopters.

Interpretation: The policy contribution is to quantify the margin on which intervention should operate. The best policy is not necessarily the most aggressive acceleration policy; it is the one that corrects the specific source of inertia without inducing excessive early adoption costs.

3 Data and measurement

3.1 Main adoption panel

- The main dataset tracks adoption of digital projectors by French movie theatres between June 2005 and April 2012.
- The panel contains theatre-screen-level adoption information: at each observation date, the number of digital projectors acquired is known for every active theatre.
- The full sample includes 1,671 theatres located in 1,169 markets and observed over 13 dates after the paper harmonizes observation timing.
- The structural analysis focuses on theatres with at least four screens, because very small theatres were subject to heavy government and regional subsidies that make their choices less informative about private adoption incentives.
- The main structural sample contains 399 theatres with at least four screens.

Interpretation: The paper studies strategic and economic adoption incentives. Excluding highly subsidized small theatres reduces contamination from externally imposed adoption timing.

3.2 Auxiliary datasets

- French National Center of Cinematography data provide theatre characteristics, addresses, number of screens and seats, owner identity, art-house status, market population, and digital movie availability.
- European Audiovisual Observatory data provide digital projector cost information.
- Box Office Mojo data identify whether U.S. movie releases were released in digital in the U.S., which the paper uses as an instrument for French digital movie availability.

Interpretation: The identification strategy requires separating theatre-side adoption from software-side movie availability. The U.S. digital release variable shifts French digital availability but is plausibly not driven by the installed base of digital screens in France.

3.3 Key variables

- **Theatre size:** total number of screens in the theatre.
- **Digital state:** number of screens converted to digital at the start of the period.
- **Adoption action:** number of additional screens converted in the period.
- **Market competition:** number of competing screens in the same local market.
- **Digital movie availability:** share of movies released in digital in France.
- **Adoption cost:** per-screen cost of digital conversion, net of virtual print fee contributions and including ancillary costs.
- **Type variables:** theatre size, market category, art-house status, chain affiliation, and local competition.

Interpretation: The key empirical variation comes from heterogeneous theatres facing common aggregate diffusion conditions: costs fall over time, digital movie availability rises, and larger theatres can gradually convert their screen stock.

3.4 Why the French movie industry is useful

- France is large enough to study industry-level diffusion but institutionally compact enough to observe most relevant theatres.
- The switch from film to digital is a clean technology-standard transition: old and new systems are incompatible.
- Theatres and distributors face complementary adoption decisions, creating the network externalities required by the theory.
- Adoption occurs screen by screen, which allows the paper to observe intensive-margin adoption within firms.

Interpretation: This is a rare setting where the hardware side, software side, costs, and adoption timing can all be linked in one structural model.

4 Model and empirical specifications

4.1 State variables and transition of theatres

There are N theatres indexed by i . A theatre type $\tau(i)$ is fixed and includes size, market characteristics, art-house status, and chain affiliation. Theatre i has $S_{\tau(i)}$ screens, and s_{it} of them are digital at the beginning of period t .

The individual state is

$$x_{it} = (\tau(i), s_{it}), \quad (1)$$

and the full industry state is

$$x_t = ((\tau(1), s_{1t}), \dots, (\tau(N), s_{Nt})). \quad (2)$$

If theatre i converts a_{it} screens in period t , its state evolves as

$$s_{i,t+1} = s_{it} + a_{it}, \quad a_{it} \leq S_{\tau(i)} - s_{it}. \quad (1)$$

Interpretation: The state variable is discrete and durable: once a screen becomes digital, it stays digital. This irreversibility makes adoption timing central.

4.2 Digital movie availability and network effects

Digital movie availability is summarized by the share of movies released in digital, h_t . It depends on the installed base of digital screens and an exogenous shifter from the U.S. market:

$$h_t = \Gamma(x_t, h_{US,t}). \quad (3)$$

Interpretation: This reduced-form distributor reaction function is the upstream side of the network effect. More digital screens make digital release more attractive; more digital movies make projectors more attractive.

4.3 Theatre payoff function

If theatre i converts a_{it} screens, its single-period payoff is

$$\Pi_i(a_{it}, x_t, p_t, h_{US,t}, \varepsilon_{it}) = \pi_{\tau(i)}(s_{it}, h_t) - a_{it}(p_t + \varepsilon_{it}). \quad (4)$$

Here p_t is the common per-screen adoption cost and ε_{it} is a private adoption shock.

The difference in current payoffs between converting a and a' screens is

$$\Delta\Pi_i(a, a', x_t, p_t, h_{US,t}, \varepsilon_i) = -(a - a')(p_t + \varepsilon_i). \quad (5)$$

Interpretation: Operating profits depend on the existing digital installed base within a theatre and on digital movie availability. Current conversion costs are linear in the number of screens converted, but dynamic incentives enter through future values.

4.4 Markov perfect equilibrium and the dimensionality problem

A full Markov perfect equilibrium would let each theatre condition on the full industry state vector x_t . With 399 theatres and at least five possible states per theatre, this state space is computationally infeasible.

The paper therefore uses a non-stationary oblivious equilibrium. Theatres condition on

$$(\tau(i), s_{it}, t) \quad (3)$$

and take the aggregate path of digital movie availability and adoption costs as deterministic.

Interpretation: The oblivious approximation keeps the aggregate network-effect channel but removes the need for each theatre to track the entire vector of competitors' states.

4.5 Non-stationary oblivious value function

For a non-stationary oblivious strategy profile, the value function is

$$V_{\tau(i),t}(s_{it}|\tilde{a}^{no}, a^{no}) = E_{\varepsilon_{it}} [\pi_{\tau(i)}(s_{it}, h_t(a^{no})) - \tilde{a}_t^{no}(x_{it}, \varepsilon_{it})(p_t + \varepsilon_{it}) + \beta V_{\tau(i),t+1}(s_{i,t+1}|\tilde{a}^{no}, a^{no})]. \quad (10)$$

Interpretation: The value function captures the dynamic tradeoff: adopt now and pay high current cost but receive digital operating benefits earlier, or wait for lower equipment costs and more complementary digital movies.

4.6 Estimation approach

The first step estimates conditional choice probabilities (CCPs) for adoption using an ordered probit. The second step chooses structural payoff parameters so that model-implied choice-specific value differences match those implied by the estimated CCPs.

The minimum-distance objective is

$$Q(\alpha) = \left\| \Delta \widehat{W}_t(a_{k+1}, a_k | \tau, s; \alpha) - \Delta \widetilde{W}_t(a_{k+1}, a_k | \tau, s) \right\|^2. \quad (19)$$

Interpretation: CCPs reveal how firms trade off current adoption costs against future benefits. The second step inverts those adoption choices into estimates of operating profits from digital conversion.

4.7 Identification of digital movie availability

The paper replaces h_t with time dummies in the adoption-profit equation and then regresses the time dummy coefficients on French digital movie availability. It instruments French digital movie availability with digital movie availability in the U.S.

Interpretation: The U.S. instrument addresses the simultaneity between French theatre adoption and French distributors' digital release decisions. It shifts software availability in France through global release technology but is less likely to be driven by French screen adoption.

5 Identification and quantitative design

5.1 What the model identifies

- The first-step CCPs identify how theatre adoption probabilities respond to time, theatre size, art-house status, current digital share, local market conditions, and chain affiliation.
- The second-step structural estimates identify the value of digital operating profits per screen and how this value changes with digital movie availability.
- Counterfactual simulations use the estimated payoff model to compare the decentralized market path with efficient benchmarks and policy interventions.

Interpretation: The estimated model turns adoption timing into information about the economic value of digital conversion.

5.2 Market outcome versus efficient benchmarks

- The **market outcome** is the non-cooperative equilibrium adoption path.
- The **planner benchmark** maximizes aggregate theatre profits taking distributors' reaction function as given.
- The **coordination benchmark** allows coordination between downstream theatres and upstream distributors.

Interpretation: The comparison separates two layers of inefficiency: adoption externalities among theatres and upstream distributor coordination failure.

5.3 Policy counterfactuals

- **Blanket subsidies:** every theatre receives a subsidy covering a fraction γ of conversion cost.
- **Targeted subsidies:** only miniplexes receive the subsidy.
- Each policy is evaluated by solving a counterfactual NOE, simulating the adoption path, and calculating aggregate theatre surplus net of adoption costs.

Interpretation: The subsidies are not merely accounting exercises. They alter dynamic incentives, adoption timing, and digital movie availability.

5.4 Main threats to interpretation

- The model treats theatre chains as making adoption decisions independently across theatres to preserve tractability.
- The distributor side is reduced form rather than a fully estimated upstream game.
- The NOE approximation abstracts from firm-specific tracking of all competitors' states.
- The structural estimates rely on observed adoption and on the assumption that U.S. digital releases are a valid shifter for French digital movie availability.
- The welfare analysis focuses on theatres' surplus, not consumer surplus or distributor profits separately.

Interpretation: These limitations affect how broadly one should interpret the welfare numbers. The strongest claim is about the measured cost of adoption delay in the French exhibition sector, not the complete social welfare of digital cinema.

6 Tables and figures

6.1 Figure 1: Data and descriptive statistics

Read:

- Panel (a) plots the diffusion of digital screens over time and marks the observation dates used in the panel.
- The digital screen share is nearly zero before 2008, rises gradually around 2008-2010, and accelerates sharply after 2010.
- Panel (b) shows that hardware adoption costs fall from about 84,000 euros per screen to under 60,000 euros over the diffusion period.
- The two panels jointly show why adoption timing is dynamic: benefits rise with digital movie availability, while costs fall over time.

Interpretation: Figure 1 gives the basic economic environment. Theatres face a real option: adopt early at higher cost and with fewer digital movies, or wait for lower prices and stronger complementary goods availability. This option is the source of gradual diffusion and the reason static adoption models would miss important timing incentives.

Economic message: The movie industry experienced both a falling-cost technology path and a growing network of complementary digital movies. The welfare question is whether private firms waited too long relative to the social optimum.

What not to over-read: The figure alone does not prove excess inertia. A slow S-shaped diffusion path may be efficient if costs are falling quickly or benefits are initially low. The structural model is needed to determine whether the timing is inefficient.

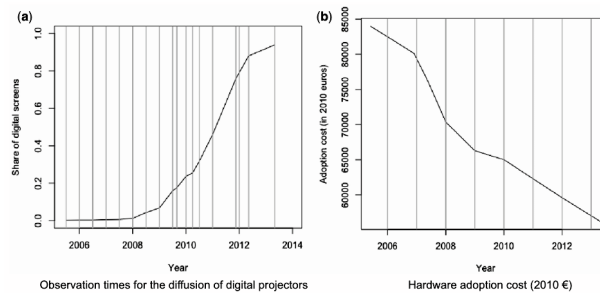


FIGURE 1
Data and descriptive statistics

Figure 1: Data and descriptive statistics.

6.2 Table 1: Summary statistics by market size and Table 2: Summary statistics

Read:

- Table 1 reports 399 theatres located in 293 markets, with an average of 8.12 screens per theatre.
- Theatres in larger urban areas have more screens and face denser local competition.
- Art-house shares are higher in smaller urban and rural areas than in Paris and large urban units.
- Table 2 reports a mean of 8.12 screens, mean art-house status of 0.34, and mean 8.70 rival screens.

- The average cost of digital conversion is about 69,862 euros per screen, with substantial time-series variation.

Interpretation: These summary statistics justify modelling theatre heterogeneity explicitly. Market size, theatre size, local competition, chain affiliation, and art-house status vary materially, so a single representative theatre would hide important differences in adoption incentives.

Economic message: The adoption decision is not uniform across firms. Larger theatres, chain theatres, and theatres in dense markets can face different costs and benefits from early adoption, which affects both the observed diffusion path and policy targeting.

What not to over-read: The tables are cross-sectional descriptions, not causal estimates. Higher screen counts in large markets do not by themselves imply faster adoption or larger digital profits; those patterns must be estimated in the dynamic model.

TABLE 1
Summary statistics by market size (theatres with at least four screens)

Geographic location	Theatres	Markets	Theatre size (mean)	Art house (share)	Screens per market (mean)	Screens per market (st.dev)
Urban unit—>100k inhab	174	101	9.17	0.21	15.79	8.43
Urban unit—20 to 100k inhab	126	116	7.02	0.56	7.63	2.99
Urban unit—<20k inhab and rural	17	17	6.65	0.65	6.65	3.55
Paris	37	15	7.19	0.14	17.73	9.57
Paris—inner suburbs	18	18	9.22	0.28	9.22	4.98
Paris—outer suburbs	27	26	7.93	0.19	8.23	4.32
National	399	293	8.12	0.34	11.05	7.26

Table 1: Summary statistics by market size.

TABLE 2
Summary statistics (theatres with at least four screens)

	Minimum	Mean	Maximum	St. deviation
Theatres characteristics				
Number of screens	4	8.12	23	3.74
Art house label	0	0.34	1	0.47
Number of rival screens	0	8.70	44	11.27
Theatres size				
Miniplexes (4–7 screens)	0	0.53	1	0.50
Multi- and Megaplexes (eight screens or more)	0	0.47	1	0.50
Theatres chains (indicator)				
UGC	0	0.08	1	0.28
Gaumont-Pathé	0	0.17	1	0.37
CGR	0	0.10	1	0.30
Cost of digital conversion (per screen, 2010 €)	56 000	69 862	84 000	10 166

Table 2: Summary statistics.

6.3 Figure 2: Within-theatre adoption

Read:

- Panel (a) shows that new digital screens increase sharply after 2009, peaking around 2012.
- Panel (b) decomposes new screens into inter-firm adoption and intra-firm adoption.
- Early adoption is mostly inter-firm: theatres install their first digital screens.
- Later adoption is increasingly intra-firm: theatres that already adopted continue converting additional screens.

Interpretation: Figure 2 shows why adoption must be modelled at the screen level. A theatre does not simply switch from old to new technology in one step. It often experiments or partially converts, then completes conversion as costs fall and digital movie availability rises.

Economic message: Partial adoption is central to the paper’s mechanism. A theatre can create network benefits for distributors and other theatres while still maintaining film capacity, so adoption externalities appear before full conversion.

What not to over-read: The inter/intra decomposition does not reveal the reason for adoption by itself. Intra-firm conversion can reflect rising benefits, falling costs, or both.

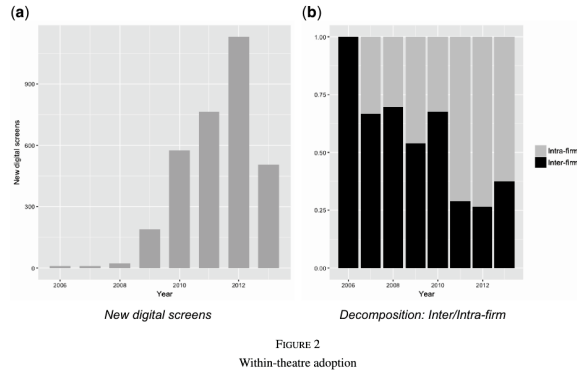


Figure 2: Within-theatre adoption.

TABLE 3
Adoption policy function

	Dependent variable: Share of screens converted a_{it}/S_{it}							
	(1)		(2)		(3)		(4)	
	Estimate	s.e	Estimate	s.e	Estimate	s.e	Estimate	s.e
Time	0.664	0.065	0.665	0.065	0.691	0.067	-0.243	0.056
Time squared	-0.021	0.003	-0.021	0.003	-0.020	0.003	0.032	0.004
Time $\times$ Art house	-0.211	0.147	-0.217	0.147	-0.291	0.147	0.129	0.009
Time squared $\times$ Art house	0.016	0.007	0.016	0.007	0.019	0.007	-0.007	0.001
Own screens	0.091	0.031	0.086	0.032	0.069	0.034	-0.219	0.043
Own screens squared	-0.003	0.001	-0.003	0.001	-0.001	0.002	-0.010	0.002
Art house	0.194	0.792	0.202	0.791	0.612	0.787	0.001	0.143
Competitors' screens (film and digital)	-0.003	0.002	-0.005	0.003	-0.001	0.003	0.012	0.006
Own share of d-screens	-1.360	0.111	-1.380	0.112	-1.666	0.118	-0.028	0.039
Market dummies								
Paris—outer suburbs			-0.239	0.170	-0.108	0.172	-0.005	0.126
Urban unit—20k–100k inhabitants			-0.086	0.142	-0.091	0.144	-0.004	0.124
Urban unit—>100k inhabitants			-0.078	0.148	-0.097	0.149	-0.010	0.118
Paris—inner suburbs			-0.345	0.189	-0.174	0.191	-0.005	0.069
Paris			-0.017	0.172	0.186	0.174	-0.004	0.079
Chain dummies								
Gaumont-Pathé					-0.151	0.080	-0.001	0.135
CGR					0.328	0.094	-0.000	0.066
UGC					-0.914	0.133	-0.007	0.046
Interactions: own screens $\times$ other variables							✓	
Observations	4,788		4,788		4,788		4,788	

Table 3: Adoption policy function.

6.4 Table 3: Adoption policy function

Read:

- The ordered probit estimates show positive time effects and negative time-squared effects in the first three specifications.
- Own screens have a positive coefficient while own screens squared is negative, suggesting larger theatres are more likely to adopt but with diminishing marginal effects.
- The share of a theatre's own screens already converted has a strongly negative coefficient. Theatres lagging behind in conversion have stronger incentives to convert additional screens.
- Competitors' screens are not significant in the adoption probability, supporting the modelling choice that strategic externalities operate mainly at the industry level rather than local market level.
- Among chain dummies, CGR is more likely to adopt and UGC less likely in the specification with chain effects.

Interpretation: Table 3 is the empirical bridge between observed adoption behavior and the structural model. It shows that adoption responds to time, theatre size, and the theatre's own conversion share, while local competitor screens are not the main driver. This supports the model's focus on aggregate network effects from digital movie availability.

Economic message: The adoption policy function demonstrates that theatres behave dynamically: they delay when they still have many options, but convert more quickly when the technology becomes broadly useful and costs fall.

What not to over-read: The coefficients are reduced-form policy-function estimates. They are not direct estimates of profits. A positive coefficient on time captures multiple forces - falling costs, rising complementary goods, and industry-wide learning - which are separated only in the structural second step.

6.5 Table 4: List of state variables

Read:

- Table 4 separates exogenous state variables, endogenous state variables, and unobserved state components.

- Observed exogenous variables include theatre type, total screens, adoption cost, U.S. digital movie availability, and the estimated payoff function.
- Endogenous observed variables include the number of converted screens, the theatre state, the industry state, and French digital movie availability.
- The firm-specific adoption shock and non-stationary oblivious belief about digital movie availability are unobserved.

Interpretation: This table clarifies what the model asks firms to condition on and what the econometrician observes. It is especially important because the NOE approximation replaces the full industry state with a deterministic aggregate path.

Economic message: The state variables encode the core economic tradeoff: current adoption costs, current digital capability, expected complementary movie supply, and future adoption options.

What not to over-read: The table is a modelling map, not a result. It does not validate the NOE approximation; it simply makes transparent what is included and what must be estimated or assumed.

	Description	Observed?
Exogenous state variables		
$\mathbf{r}(t)$	Firm i 's type (screens $S_{i(t)}$, market characteristics, art house, chain)	Observed
$S_{i(t)}$	Firm i 's total number of screens	Observed
p_i	Per-screen adoption cost	Observed
ϵ_{it}	Firm-specific private adoption shock	No
$h_{t,US}$	Share of movies released in digital in t in the US	Observed
$\pi_{i(t)}(s_{it}, h_t)$	Firm i 's single-period payoff function	Estimated
Endogenous state variables		
a_{it}	Number of screens converted in t	Observed
s_{it}	Number of screens at the start of period t	Observed
$\mathbf{x}_{it}$	$(\mathbf{r}(t), s_{it})$	Observed
$\mathbf{x}_t$	$(\mathbf{x}_{1t}, \dots, \mathbf{x}_{N_t}, \dots, \mathbf{x}_{N_t})$	Observed
h_t	Share of movies released in digital in t in France	Observed
$\tilde{h}_t(\mathbf{a}^{no})$	Non-stationary oblivious belief about h_t ($E_{\mathbf{a}}[h_t \mathbf{x}_t]$) given strategy profile $\mathbf{a}^{no}$	No

Table 4: List of state variables.

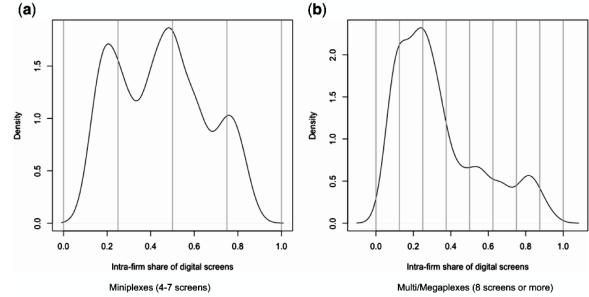


Figure 3: Density estimate of intra-firm rate of adoption by firm size.

6.6 Figure 3: Density estimate of intra-firm rate of adoption by firm size

Read:

- Panel (a) shows that miniplexes display a multi-peaked distribution of within-firm digital shares.
- Panel (b) shows that multi/megaplexes are more concentrated at relatively low digital shares, with a long right tail.
- Both panels condition on theatres that have partially adopted: the share is greater than zero and less than one.
- Larger theatres have more possibilities for partial adoption because they have more screens over which to spread conversion.

Interpretation: Figure 3 motivates discretizing adoption choices differently for miniplexes and multi/megaplexes. Larger theatres can convert one or two screens at a time and wait, whereas smaller theatres have coarser adoption steps.

Economic message: The distribution of partial adoption is evidence that the technology transition is not simply an irreversible all-or-nothing investment. Gradual conversion creates both flexibility and delay.

What not to over-read: The density plot is conditional on partial adopters, so it does not describe the whole industry. It should not be interpreted as the unconditional distribution of digital conversion.

6.7 Table 5: Structural parameter estimates

Read:

- The baseline profit per digital screen is estimated at about 4,025 euros per six-month period.
- Profits per digital screen increase with own screens, consistent with scale economies in theatre operation.
- CGR theatres have higher estimated digital-screen profits, while UGC theatres have lower estimated digital-screen profits.
- The decay term is strongly decreasing in the share of digital movies: as more movies become available in digital, the opportunity cost of being digital falls and the value of a digital screen rises.
- The estimated magnitudes are consistent with labour cost savings from replacing film projection with digital projection.

Interpretation: Table 5 converts observed adoption choices into economic profit estimates. The estimates imply that digital conversion creates meaningful cost reductions, but the value depends on theatre size, chain affiliation, and digital movie availability.

Economic message: Digital adoption is profitable mainly because it reduces operating costs and becomes more useful as digital movies become widely available. The profit heterogeneity explains why some theatres adopt early while others wait.

What not to over-read: These parameters are structural estimates conditional on the model and on the normalization that film profits are zero. They should not be read as directly observed accounting profits.

	Estimate	s.e.
Profit per digital screen: $\pi_d(\tau_i)$		
Constant	4,024.86	959.18
Own screens	112.00	51.17
Art house	638.23	696.07
Competitors' screens	-2.16	5.42
Market dummies		
Paris—outer suburbs	-219.65	278.49
Urban unit—20k–100k inhabitants	-150.87	223.03
Urban unit—>100k inhabitants	-177.61	230.50
Paris—inner suburbs	-325.05	310.92
Paris	246.76	295.56
Chain dummies		
Gaumont-Pathé	-229.05	146.82
CGR	622.02	166.90
UGC	-1,650.32	430.57
Decay function: $\delta(\tau_i, h_i)$		
Constant	3,920.84	1,096.80
Own screens	17.03	42.71

Table 5: Structural parameter estimates.

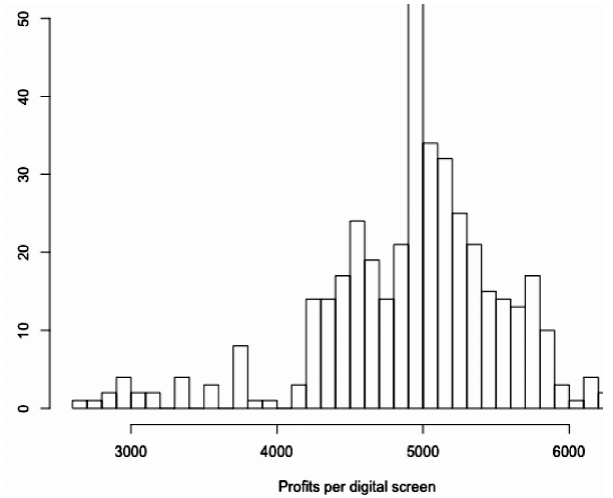


FIGURE 4

Figure 4: Histogram of profit per digital screen.

6.8 Figure 4: Histogram of profit per digital screen

Read:

- Figure 4 plots the distribution of estimated single-period profits per digital screen across theatres.
- The distribution is concentrated around 4,500-5,500 euros, with a mean around 4,928 euros and a median around 4,986 euros in the paper's discussion.
- The distribution has some lower-profit theatres, but most theatres have positive estimated gains from digital conversion.

Interpretation: The histogram translates Table 5's parameter estimates into theatre-level heterogeneity. Most theatres benefit from digital conversion, but the timing still varies because adoption costs are high and complementary digital movies are initially scarce.

Economic message: The cost-of-standardization problem is not that the new technology is unprofitable. It is that a profitable technology can still be adopted too slowly when benefits depend on others adopting and when complementary goods are slow to switch.

What not to over-read: The histogram does not measure total profits from theatres; it is a model-implied relative profit per digital screen relative to film.

6.9 Figure 5: Counterfactual diffusion paths under efficient benchmarks and subsidies

Read:

- The top panels compare the market outcome with the planner and coordination benchmarks.
- Both efficient benchmarks diffuse faster than the market outcome, especially in the early transition years.
- The coordination benchmark moves faster than the planner benchmark because it also resolves upstream distributor coordination failure.
- The middle panels show blanket subsidies. Larger subsidies speed adoption, but high subsidies can generate excessive early adoption costs.
- The bottom panels show targeted subsidies to miniplexes. Targeting smaller theatres can accelerate diffusion while using less public spending.

Interpretation: Figure 5 is the central counterfactual figure. It demonstrates that the market transition is slower than efficient benchmarks, but it also shows that the policy problem is not simply "speed up adoption as much as possible." Overly aggressive subsidies overshoot the optimal timing.

Economic message: The figure illustrates the two margins of policy design: correct the externality and avoid excessive early conversion. Targeting can improve efficiency because the adoption delay is concentrated among smaller theatres.

What not to over-read: The figure shows diffusion paths, not welfare directly. Welfare must include adoption costs, which is why Tables 6-8 are needed.

6.10 Table 6: Theatres' surplus under the three scenarios

Read:

- Net industry profits are 76.18 million euros under the market outcome.

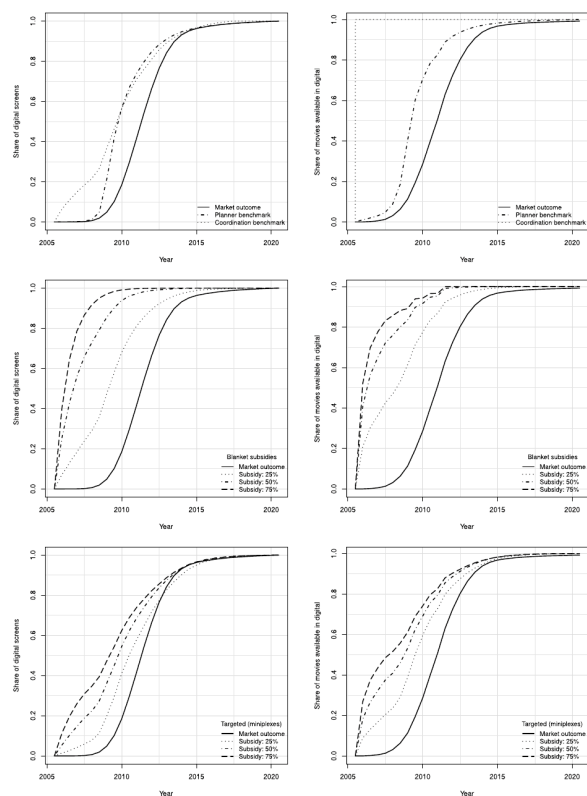


Figure 5: Counterfactual diffusion paths under efficient benchmarks, blanket subsidies, and targeted subsidies.

TABLE 6
Theatres' surplus under the three scenarios (in millions, 2010 €)

	Market outcome	Planner benchmark	Coordination benchmark
Gross industry profits	167.54	193.75	209.52
Adoption costs	91.36	111.49	118.88
Net industry profits	76.18	82.25	90.65
Change in net profits	0	6.07	14.46
Change in net profits (in %)	0	7.97	18.98

Notes: Profits from film are normalized to 0; all profits are relative to the status-quo with no adoption.

Table 6: Theatres' surplus under the market, planner, and coordination scenarios.

- Net industry profits rise to 82.25 million euros under the planner benchmark, a gain of 6.07 million euros or 7.97%.
- Net industry profits rise to 90.65 million euros under the coordination benchmark, a gain of 14.46 million euros or 18.98%.
- Adoption costs are higher under the efficient benchmarks because adoption occurs earlier and more intensively, but gross profits rise enough to justify those costs.

Interpretation: Table 6 quantifies excess inertia. The planner benchmark isolates downstream adoption externalities among theatres, while the coordination benchmark adds the value of resolving upstream distributor coordination problems. The difference between the two benchmarks measures the upstream component.

Economic message: The market leaves meaningful surplus on the table. The delay is not merely an accounting feature of slower adoption; it lowers aggregate theatre surplus net of adoption costs.

What not to over-read: These surplus numbers are theatre surplus relative to the no-adoption film benchmark. They are not total social welfare including consumer surplus and distributor profits.

6.11 Table 7: Theatres' surplus under counterfactual blanket subsidies

Read:

- A 25% blanket subsidy raises net industry profits from 76.18 to 82.22 million euros, nearly matching the planner benchmark.
- A 50% blanket subsidy lowers net industry profits to 70.56 million euros.
- A 75% blanket subsidy lowers net industry profits further to 60.19 million euros.
- High blanket subsidies increase gross profits, but they also induce excessive adoption costs.

Interpretation: Table 7 shows that subsidy design is non-monotone. A modest subsidy corrects externalities and improves surplus. Larger subsidies force adoption too early relative to costs, so they reduce net surplus.

Economic message: The policy lesson is subtle: when technologies improve and costs fall over time, the efficient policy is not to eliminate all delay. Some waiting is socially efficient.

What not to over-read: The table is not an argument against subsidies in general. It shows that the level and targeting of subsidies matter.

	Market outcome	Blanket subsidies		
		$\gamma=0.25$	$\gamma=0.50$	$\gamma=0.75$
Gross industry profits	167.54	217.95	262.06	280.74
Adoption costs	91.36	135.73	191.5	220.55
Net industry profits	76.18	82.22	70.56	60.19
Change in net profits	0	6.04	-5.63	-15.99
Change in net profits (in %)	0	7.93	-7.38	-20.99

Notes: Profits from film are normalized to 0: all profits are relative to the status-quo with no adoption. γp_i corresponds to the subsidy level incurred by the social planner and $(1-\gamma)p_i$ is incurred by the theatre. "Adoption costs" include costs to theatres and the social planner.

diffusion paths under various broad-based subsidies ($\gamma \in \{0, 0.25, 0.5, 0.75\}$) for digital cinema

Table 7: Theatres' surplus under counterfactual blanket subsidies.

TABLE 8 Theatres' surplus under counterfactual targeted subsidies (in millions, 2010 €)				
	Market outcome	Subsidies to miniplexes		
		$\gamma=0.25$	$\gamma=0.50$	$\gamma=0.75$
Gross industry profits	167.54	184.52	201.01	212.07
Adoption costs	91.36	102.51	122.03	138.55
Net industry profits	76.18	82.01	78.97	73.52
Change in net profits	0	5.83	2.79	-2.66
Change in net profits (in %)	0	7.65	3.66	-3.49

Notes: Profits from film are normalized to 0: all profits are relative to the status-quo with no adoption. γp_i corresponds to the subsidy level incurred by the social planner and $(1-\gamma)p_i$ is incurred by the theatre. "Adoption costs" include costs to theatres and the social planner.

Table 8: Theatres' surplus under counterfactual targeted subsidies.

6.12 Table 8: Theatres' surplus under counterfactual targeted subsidies

Read:

- A 25% subsidy targeted to miniplexes raises net industry profits from 76.18 to 82.01 million euros, a 7.65% increase.
- A 50% targeted subsidy still raises net industry profits by 3.66%, unlike the corresponding blanket subsidy.
- A 75% targeted subsidy slightly reduces net industry profits by 3.49%.
- Targeting smaller theatres achieves gains close to the best blanket subsidy but at lower fiscal cost.

Interpretation: Table 8 shows that the externality problem is concentrated among smaller theatres that delay adoption. Subsidizing those theatres speeds up the slow side of the network without paying large subsidies to large early adopters.

Economic message: Targeted policies can be more cost-effective than universal ones because the welfare-relevant distortion is not evenly distributed across firms.

What not to over-read: The result is specific to this industry and targeting margin. In other standard-setting industries, the delayed side of the market may be different.

7 Short summary

- The paper studies the transition from 35mm film to digital cinema in France.
- The transition is a hardware-software standard switch with indirect network effects.
- Theatres benefit from more digital movies; distributors benefit from more digital screens.
- The market can delay switching because individual adopters do not internalize all benefits they create for other firms and for distributors.
- The paper uses detailed theatre-screen adoption data, adoption costs, theatre characteristics, and digital movie availability.
- The structural model is a dynamic game with durable screen conversion decisions.
- A non-stationary oblivious equilibrium approximation makes the dynamic game estimable despite a large state space.
- First-step CCP estimates show that adoption rises over time, larger theatres adopt faster, and theatres with fewer own digital screens are more likely to convert additional screens.
- Structural estimates imply positive per-screen digital profits consistent with labour cost savings.
- Counterfactual simulations show that market diffusion is slower than efficient benchmarks.
- Theatre surplus is 76.18 million euros under the market outcome, 82.25 million euros under the planner benchmark, and 90.65 million euros under the coordination benchmark.
- Excess inertia reduces surplus by 16% relative to full coordination.
- Downstream adoption externalities explain 41% of the surplus loss; upstream distributor coordination frictions explain the remaining 59%.
- A 25% blanket subsidy or a 25% targeted subsidy to miniplexes can substantially reduce the welfare gap.
- Higher subsidies can be inefficient because they accelerate adoption too much relative to falling costs and rising benefits.

8 Expanded conclusion

8.1 Core conclusion

Caoui's paper shows that standardization can create a real cost when a superior but incompatible technology arrives. The French movie industry eventually converged to digital cinema, but the decentralized market adopted more slowly than efficient benchmarks. The reason is not simple irrational delay: the market faces a hardware-software coordination problem in which individual theatres do not internalize how their digital projectors encourage distributors to release digital movies and benefit other theatres.

8.2 Economic intuition

The central intuition is a dynamic chicken-and-egg problem. Theatres want digital movies before they install digital projectors; distributors want digital screens before they release digital movies. Falling hardware costs make waiting privately attractive, while network externalities make early adoption socially valuable. This combination creates excess inertia: private adoption timing is too late relative to a planner who internalizes the network spillovers.

8.3 Why the welfare decomposition matters

The distinction between the planner and coordination benchmarks is important. The planner benchmark corrects downstream externalities among theatres while holding distributors' reaction function fixed. The coordination benchmark also resolves upstream distributor coordination failure. This allows the paper to assign 41% of the surplus loss to downstream externalities and 59% to upstream inertia. Without this decomposition, the paper would only show that diffusion is slow, not why it is slow.

8.4 Policy implications

The policy results are nuanced. Subsidies can help, but only if calibrated to the source of the distortion. A modest blanket subsidy raises net profits, but larger subsidies accelerate adoption too much. A targeted subsidy to miniplexes can achieve a similar surplus gain at a lower fiscal cost because smaller theatres are the slow adopters that help close the network externality gap. The broader lesson is that policy in standards markets should identify the delayed side of the network and target that margin.

8.5 Methodological contribution

The paper provides a useful template for studying large dynamic network industries. It combines detailed adoption data, a dynamic game, a reduced-form complementary-goods supply response, and non-stationary oblivious equilibrium. This makes it possible to estimate counterfactual adoption paths in an industry with hundreds of firms, where full Markov perfect equilibrium methods would be computationally infeasible.

8.6 External validity

The mechanism applies beyond movie theatres. Similar standardization problems arise in electric vehicles and charging networks, payment technologies, operating systems, health data formats, broadband infrastructure, and communication standards. In each case, private adoption depends on complementary goods and other adopters. However, the specific policy prescription depends on where the slow side of the network lies and whether targeted subsidies or coordination mandates are feasible.

8.7 Limitations and open questions

The welfare analysis focuses on theatre surplus, not the full distribution of gains across consumers, distributors, hardware suppliers, and policy makers. Distributor behavior is modelled through a reduced-form reaction function rather than a full upstream dynamic game. Chain-level coordination across theatres is simplified. Future work could extend the framework to measure consumer surplus, model upstream strategic behavior more explicitly, and study whether mandates or standard-setting institutions outperform subsidies in different industries.

8.8 Takeaway

The paper’s main takeaway is that standards are double-edged. They create value by coordinating users on compatible technologies, but they can also slow the transition to better incompatible technologies. Quantifying that delay requires a dynamic model of adoption and complementary-good supply. In the French movie industry, the cost of standardization is large enough to justify carefully targeted intervention.